



Stained Glass Panels

Hackathon Grasshopper Users Meeting

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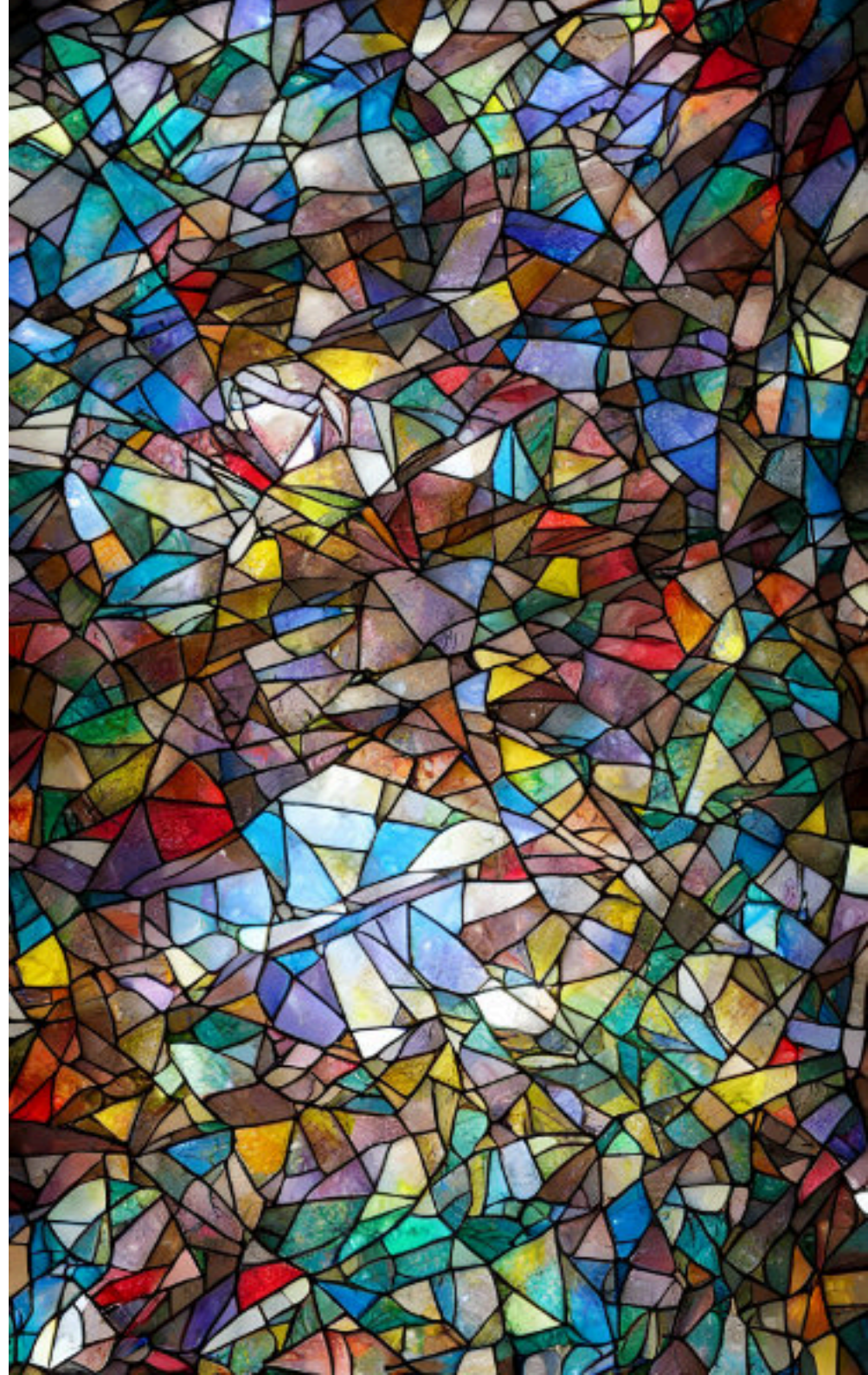
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The Problem

We want unique and beautiful elements in our spaces.

Problem: Custom ornament is expensive because it requires intense manual design and specialized craftsmanship.

Result: Budget constraints force us back into sterile, mass-produced minimalism.



Our Solution

We move the complexity from manual labor to software.

Platform: A configurator that lets anyone turn an image into a stained glass panel.

Output: Instant, 3D-printable files. No specialized craft skills required; no fortune needed.

The Inputs



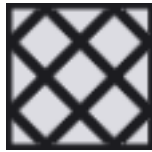
Raster



Number of colours



Colours



Pattern of the panels

The Outputs

Visualisation

Model 3D in .glb

The workflow

Parallel tracks from the start. We split the team into:

- image-processing - Somia, Mateusz
- geometry - Martyna, Ewa
- web/render/deploy - Bartłomiej
- UI and UX design - Martyna

The Tools



Rhino 7 + Grasshopper

First geometry engine — image sampling, colour quantization, panelization, BREP frames/extrusion



Python / Jupyter

Early model transport from Grasshopper to the web viewer



Rust + wasm-bindgen

The production engine, compiled to WebAssembly (sg-wasm/)



three.js (WebGPU)

The 3D "beauty" renderer



Illustrator/Photoshop

Logo (star rosette), grasshopper mascot, loading GIF, app mockups



Cloudflare + Wrangler

Hosting (no-build static site)



STAINED GLASS

Pick your colours. Pick your pattern. Let the light do the rest.

Stained glass used to be the language of cathedrals and grand estates.
We're bringing it back — to your home, your studio, your space. Design
your own piece, choose your colours and pattern, and we'll print it. No
craft skills needed. No fortune required.

HOW IT WORKS

MAKE MY ART



Grashoppers Users Meeting Warsaw

English ^



What's Next?

Our prototype can develop in multiple ways, we propose this roadmap to advance our Idea:

- 1) Create viable way of connecting panels to each other using Printable frame exported using website
- 2) Line recognition of pictures uploaded on website
- 3) Optimised technical view with detail of connections and 2d drawings
- 4) The printer selection
- 5) Estimated price of the finished product
- 6) Gallery of the pictures with finished model
- 7) Online shop for special orders and infrastructure
- 8) 3D printer farms for orders
- 9) Metal printing of the mesh

Thank you!